Lecture 4

Entity-Relationship to Relational Mapping

CSC1022 Introduction to Databases

CSC1005 Data Science & Databases
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Agenda

1. Introduction
2. Step 1: Mapping of Regular Entity Types
3. Step 2: Mapping of Weak Entity Types
4. Step 3: Mapping of Binary 1:1 Relationship Types
5. Step 4. Mapping of Binary 1:N Relationship Types
6. Step 5. Mapping of Binary M:N Relationship Types
7. Step 6. Mapping of Multivalued Attributes
8. Step 7. Mapping of N-ary Relationship Types.

Learning Objectives

- How do we convert from a conceptual (E-R) schema to a Relational (MySQL) schema?
- After E-R modelling, the next step is the *logical database design* or *data model mapping* step.
- We need a *methodology* to create a **Relational** schema from an **Entity-Relationship** (ER) schema.
- Tools exist which use ER diagrams to develop the schema graphically and automatically concert it into a relational database schema.
- *For Reference, you can follow this video, which takes a similar example of our company ER Model:*

  https://youtu.be/xQRRf5fOAt8?si=Ky0QLxhzeI6y_96R

Case Study: COMPANY database

- The COMPANY database example illustrates the mapping procedure (see Figure 1).
- We describe the steps of an algorithm for ER-to-Relational mapping.
- We assume that the mapping will create tables with simple *single-valued attributes*.
- The relational model constraints which include primary keys, unique keys (if any), and referential integrity constraints on the relations, will also be specified in the mapping results.

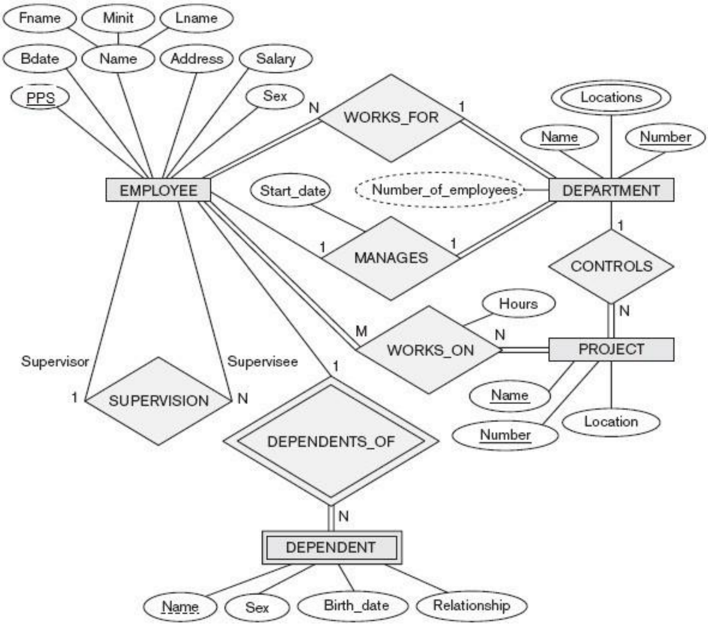

Figure 1: ER Conceptual schema for the COMPANY database

Step 1. Mapping Regular Entity Types

1. For each regular (strong) entity type *E* in the E-R schema, create a relation *R* that includes all simple attributes of *E*.
2. Include *only* the simple *component* attributes of a composite attribute.
3. Choose one of the key attributes of *E* as the primary key for *R*.
4. If the chosen key of *E* is a composite, its set of simple attributes will together form the primary key of *R*.

Mapping Keys

- If multiple keys were identified for *E*, each additional key is recorded.
- In this example, create the relations EMPLOYEE, DEPARTMENT, and PROJECT in Figure 9 to correspond to the regular entity types EMPLOYEE, DEPARTMENT, and PROJECT.
- The foreign key and relationship attributes will be added during subsequent steps.

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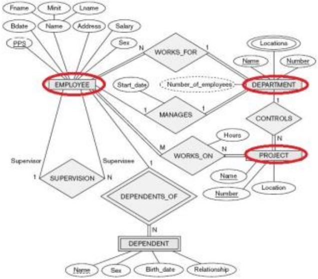
Mappings for Step 1

- Choose `PPS`, `Dnumber`, and `Pnumber` as primary keys for relations EMPLOYEE, DEPARTMENT, and PROJECT, respectively.
- Knowledge that `Dname` of DEPARTMENT and `Pname` of PROJECT are secondary keys should be kept for possible use later.
- Relations created from the mapping of entity types are called **entity relations** because each tuple represents an *entity instance*.

End of Step 1



EMPLOYEE

Fname	Minit	Lname	PPS	Bdate	Address	Sex	Salary	Dno

DEPARTMENT

Dname	Dnumber	Mgr_start_date

PROJECT

Pname	Pnumber		Dnum

Step 2. Mapping of Weak Entity Types

Inputs: *E*,the strong entity; *W*,the weak entity.

- For each weak entity type *W* in the E-R schema with owner entity type *E*:
  - create a relation *R*;
  - include all simple attributes (or simple components of composite attributes) of *W* as attributes of *R*.
- Include as foreign key attributes of *R*:
  the primary key attribute(s) of the relation(s) that correspond to the owner entity type(s).
- This takes care of mapping the identifying relationship type of *W*.

Primary Key

- The primary key of *R* is the combination of the primary key(s) of the owner(s) and the partial key of the weak entity type *W*, if any.
- If there is a weak entity type *E2* whose owner is also a weak entity type *E1*, then *E1* should be mapped before *E2* to determine its primary key *first*.

Mappings for Step 2

- Create the relation DEPENDENT to correspond to the weak entity type DEPENDENT.
- Include the primary key `PPS` from EMPLOYEE (owner entity type), as a foreign key attribute of DEPENDENT; rename it `EMP_PPS`.
- The primary key of the DEPENDENT relation is the combination {`EMP_PPS, Dependent_name`}, because Dependent_name is the partial key of DEPENDENT.

End of Step 2

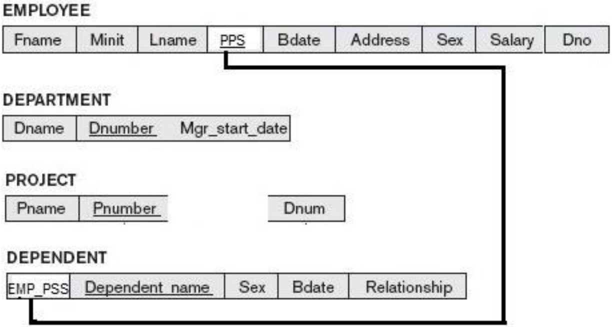


Figure 3: Weak Entities become Tables with Relationship to Strong Entities

Step 3. Mapping of Binary 1:1 Relationship Types

- For each binary 1:1 relationship type *R* in the E-R schema, identify the relations *S* and *T* that correspond to the entity types participating in *R*.
- There are three possible approaches:
  (1) the **foreign key** approach,
  (2) the **merged relationship** approach, and
  (3) the **cross-reference** or relationship relation approach.
- The first approach is the most useful and should be followed unless special conditions exist.

1. Foreign key Approach

- Choose one of the relations, say *S*, and include as a foreign key in *S* the primary key of *T*.
- It is better to choose an entity type with total participation in *R* in the role of *S*.
- Include all the simple attributes (or simple components of composite attributes) of the 1:1 relationship type *R* as attributes of *S*.

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Foreign key Approach Mappings

- Map the 1:1 relationship type MANAGES (figure 1) by choosing the participating entity type DEPARTMENT to serve in the role of *S* because its participation in the MANAGES relationship type is **total** (*every* department has a manager).
- Include the primary key of the EMPLOYEE relation as foreign key in the DEPARTMENT relation and rename it `Mgr_PPS`.
- Include the simple attribute `Start_date` of the MANAGES relationship type in the DEPARTMENT relation and rename it `Mgr_start_date` (Figure 9).

Issues with FK Approach

- Note that it is possible to include the primary key of *S* as a foreign key in *T* instead.
- This amounts to having a foreign key attribute, say `Department_managed` in the EMPLOYEE relation, but it will have a NULL value for employee tuples who do not manage a department.
- If only 2 percent of employees manage a department, then 98 percent of the foreign keys would be NULL in this case.
- Another possibility is to have foreign keys in both relations *S* and *T* redundantly, but this creates redundancy and incurs a penalty for consistency maintenance.

2. Merged Relation Approach

- An alternative mapping of a 1:1 relationship type is to merge the two entity types and the relationship into a single relation.
- This is possible when **both participations** are *total*, as this would indicate that the two tables will have the exact same number of tuples at all times.

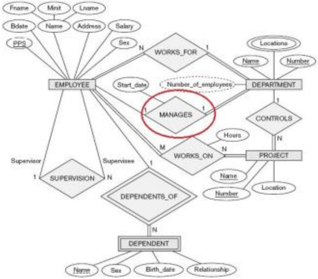
3. Cross-reference or Relationship Relation Approach

- This option creates a third relation *R* for the purpose of cross-referencing the primary keys of the two relations *S* and *T* representing the entity types.
- This approach is required for *M:N* relationships.
- The relation *R* is called a **relationship relation** as each tuple in *R* represents a relationship instance that relates one tuple from *S* with one tuple from *T*.

Cross-Reference Mappings

- The relation *R* will include the primary key attributes of *S* and *T* as foreign keys to *S* and *T*.
- The primary key of *R* will be one of the two foreign keys, and the other foreign key will be a unique key of *R*.
- The drawback is having an extra relation, and requiring an extra join operation when combining related tuples from the tables.

End of Step 3



EMPLOYEE

Fname	Minit	Lname	PPS	Bdate	Address	Sex	Salary	Dno

DEPARTMENT

Dname	Dnumber	MGR_PPS	Mgr_start_date

PROJECT

Pname	Pnumber	Dnum

DEPENDENT

EMP_PSS	Dependent_name	Sex	Bdate	Relationship

Figure 4: 1-1 Relationships

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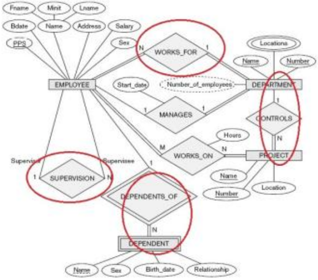
4: Mapping of Binary 1:N Relationship Types

- For each regular binary *1:N* relationship type *R*, identify the relation *S* that represents the participating entity type at the *N*-side of the relationship type.
- Include as foreign key in *S* the primary key of the relation *T* that represents the other entity type participating in *R*.
- Use this approach, as each entity instance on the *N*-side is related to *at most one* entity instance on the 1-side of the relationship type.
- Include any simple attributes (or simple components of composite attributes) of the *1:N* relationship type as attributes of *S*.

Mappings for Step 4

- Map the *1:N* relationship types WORKS_FOR, CONTROLS, and SUPERVISION from Figure 1.
- For WORKS_FOR, we include the primary key `Dnumber` of the DEPARTMENT relation as foreign key in the EMPLOYEE relation and call it `Dno`.
- For SUPERVISION, we include the primary key of the EMPLOYEE relation as foreign key in the EMPLOYEE relation itself (it's recursive) and call it `Super_PPS`.
- The CONTROLS relationship is mapped to the foreign key attribute `Dnum` of PROJECT, which references the primary key `Dnumber` of the DEPARTMENT relation.

End of Step 4



EMPLOYEE

Fname	Minit	Lname	PPS	Bdate	Address	Sex	Salary	Super_PPS	Dno

DEPARTMENT

Dname	Dnumber	MGR_PPS	Mgr_start_date

DEPT_LOCATIONS

Dnumber	Dlocation

PROJECT

Pname	Pnumber	Dnum

DEPENDENT

EMP_PSS	Dependent_name	Sex	Bdate	Relationship

Step 5: Mapping of Binary M:N Relationship Types

- For each binary *M:N* relationship type *R*, create a new relation S to represent *R*.
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types;
- The combination will form the primary key of *S*.
- Also include any simple attributes of the *M:N* relationship type (or simple components of composite attributes) as attributes of *S*.

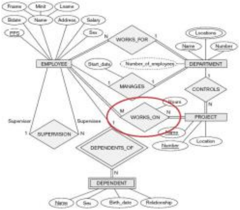
Always requires a New Table

- Note that it is not possible to represent an *M:N* relationship type by a single foreign key attribute in one of the participating relations (as with 1:1 or *1:N* relationship types) because of the *M:N* cardinality ratio.
- It is always necessary create a separate relationship relation *S*.

Mappings for Step 5

- In the case study, map the *M:N* relationship type WORKS_ON (Fig 1) by creating the relation WORKS_ON in Figure 9.
- Include the primary keys of the PROJECT and EMPLOYEE relations as foreign keys in WORKS_ON and rename them `Pno` and `EMP_PPS`, respectively.
- Include an attribute `Hours` in WORKS_ON to represent the `Hours` attribute of the relationship type.
- The primary key of the WORKS_ON relation is the combination of the foreign key attributes {`EMP_PPS`, `Pno`}.

End of Step 5



EMPLOYEE

Fname	Minit	Lname	PPS	Bdate	Address	Sex	Salary	Super_PPS	Dno

DEPARTMENT

Dname	Dnumber	MGR_PPS	Mgr_start_date

PROJECT

Pname	Pnumber	Plocation	Dnum

WORKS_ON

EMP_PSS	Pno	Hours

DEPENDENT

EMP_PSS	Dependent_name	Sex	Bdate	Relationship

Figure 6: M-N Relationships

Entity-Relationship to Relational Mapping

Introduction

Step 1: Mapping of Regular Entity Types

Step 2: Mapping of Weak Entity Types

Step 3: Mapping of Binary 1:1 Relationship Types

Step 4. Mapping of Binary 1:N Relationship Types

Step 5. Mapping of Binary M:N Relationship Types

Step 6. Mapping of Multivalued Attributes

Step 7. Mapping of N-ary Relationship Types.

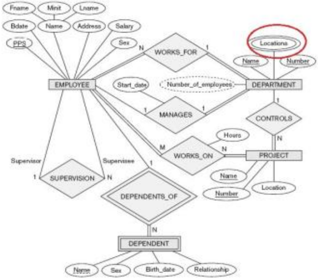
Step 6: Mapping of Multivalued Attributes

- For each multivalued attribute *A*, create a new relation *R*.
- This relation *R* will include an attribute corresponding to *A*, plus the primary key attribute *K* (as a foreign key in *R*) of the relation that represents the entity type or relationship type that has *A* as a multivalued attribute.
- The primary key of *R* is the combination of *A* and *K*.
- If the multivalued attribute is composite, include its simple components.

Mappings for Step 6

- Create a relation DEPT_LOCATIONS.
- The attribute `Dlocation` represents the multivalued attribute LOCATIONS of DEPARTMENT; `Dnumber` (as foreign key) represents the primary key of the DEPARTMENT relation.
- The primary key of DEPT_LOCATIONS is the combination of {`Dnumber, Dlocation`}.
- A separate row will exist in DEPT_LOCATIONS for each department's location.

End of Step 6



EMPLOYEE

Fname	Minit	Lname	PPS	Bdate	Address	Sex	Salary	Super_PPS	Dno

DEPARTMENT

Dname	Dnumber	MGR_PPS	Mgr_start_date

DEPT_LOCATIONS

Dnumber	Dlocation

PROJECT

Pname	Pnumber	Plocation	Dnum

WORKS_ON

EMP_PSS	Pno	Hours

DEPENDENT

EMP_PSS	Dependent_name	Sex	Bdate	Relationship

Final Step

- Figure 9 showed the COMPANY relational database schema obtained with steps 1 through 6.
- There is no mapping of *n*-ary relationship types (n > 2) because none exist in figure 1.
- These are mapped in a similar way to *M:N* relationship types by including the following additional step in the mapping algorithm.

Step 7: Mapping of *N*-ary Relationship Types.

- For each *n*-ary relationship type *R*, where n > 2, create a new relation *S* to represent *R*.
- Include as foreign key attributes in *S* the primary keys of the relations that represent the participating entity types.
- Include any simple attributes of the *n*-ary relationship type (or simple components of composite attributes) as attributes of *S*.
- The primary key of *S* is usually a combination of all the foreign keys that reference the relations representing the participating entity types.

N-ary Relationship Mappings

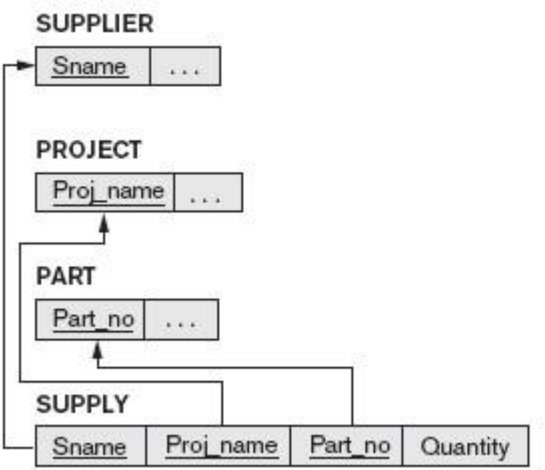


Figure 8: Mapping the n-ary relationship type SUPPLY

Final Relational Schema

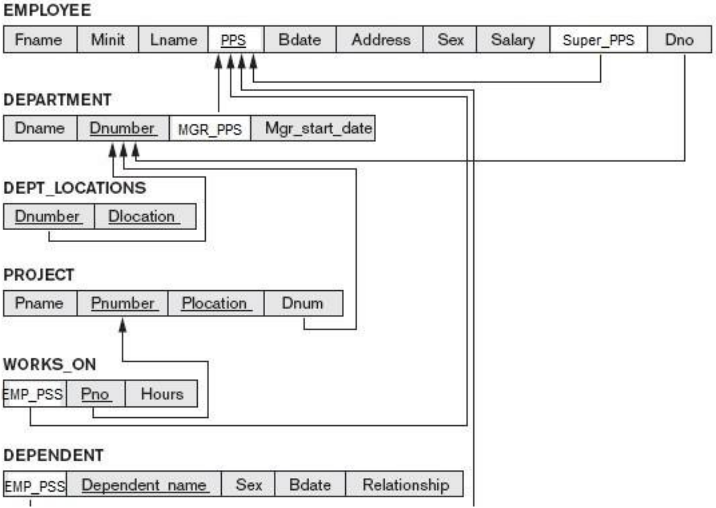


Figure 9: Result of Mapping E-R schema to relational schema

Lecture Ends . . .